

DLD QUIZ 6-B

① 3 bit comparator expressions

$$(A=B) = (A_0 \odot B_0)(A_1 \odot B_1)(A_2 \odot B_2)$$

$$(A < B) = \overline{A_2}B_2 + \overline{A_1}B_1(A_2 \odot B_2) + \overline{A_0}B_0(A_2 \odot B_2)(A_1 \odot B_1)$$

$$(A > B) = A_2\overline{B_2} + A_1\overline{B_1}(A_2 \odot B_2) + A_0\overline{B_0}(A_2 \odot B_2)(A_1 \odot B_1)$$

② 2x4 Decoder Expressions

E_1	$\overline{E_0}$	I_0	I_1	$\overline{Q_0}$	$\overline{Q_1}$	$\overline{Q_2}$	$\overline{Q_3}$
1	1	0	0	$\overline{Q_0} = \overline{E_1 \overline{E_0} I_0 I_1}$			
1	1	0	1	$\overline{Q_1} = \overline{E_1 \overline{E_0} I_0 I_1}$			
1	1	1	0	$\overline{Q_2} = \overline{E_1 \overline{E_0} I_0 I_1}$			
1	1	1	1	$\overline{Q_3} = \overline{E_1 \overline{E_0} I_0 I_1}$			

③ 4x1 MUX Function Implementation $\Sigma(1, 2, 6, 7)$

X	Y	Z	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

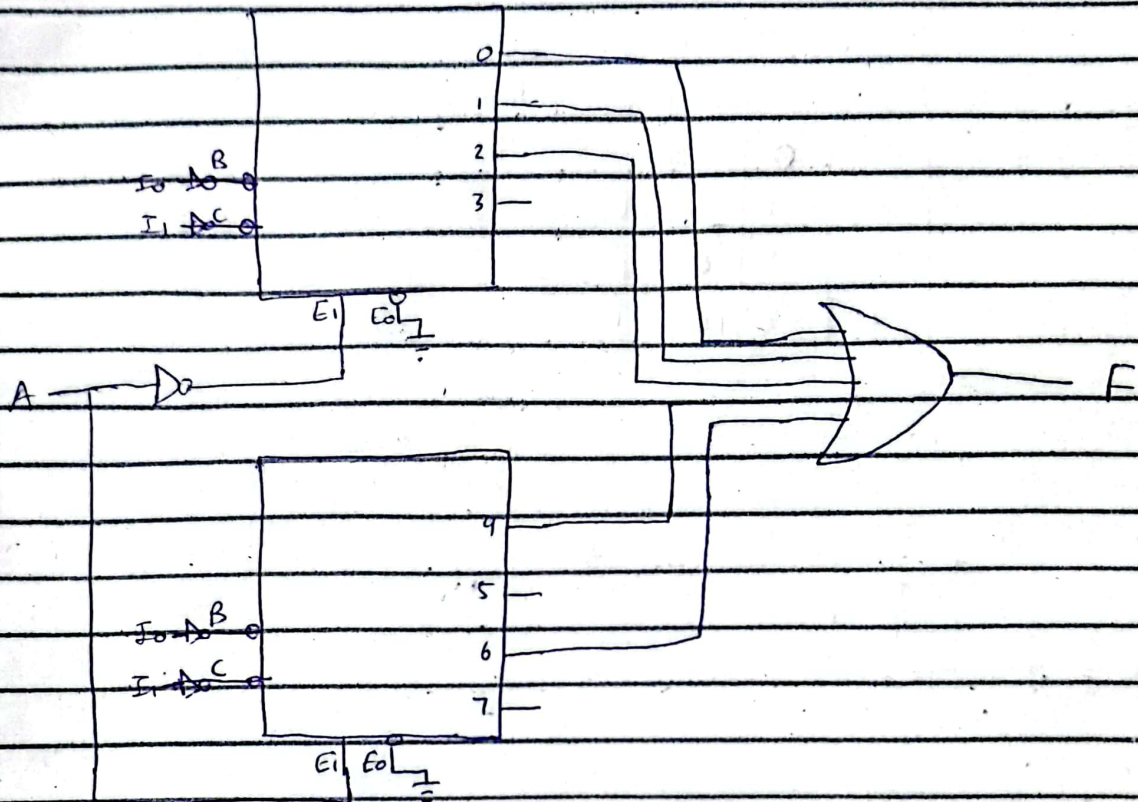
$\left. \begin{matrix} F=Z \\ F=\overline{Z} \\ F=0 \\ F=1 \end{matrix} \right\} \Sigma(1, 2, 6, 7)$

If X used as input

$\overline{X}$	0	①	②	3
X	4	5	⑥	⑦
0	$\overline{X}$	1	X	

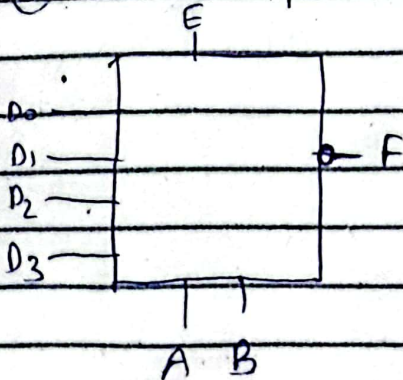
(B)

⑤ 2x4 Decoders Function Implementation $\Sigma(0,1,2,4,6)$



Connecting maxterm outputs to ~~NAND~~ ^{NAND} is also valid (but not recommended)

⑥ 4x1 MUX Expression



A	B	F
0	0	D ₀
0	1	D ₁
1	0	D ₂
1	1	D ₃

Expression:-

$$F = (E \bar{A} \bar{B} D_0 + E \bar{A} B D_1 + E A \bar{B} D_2 + E A B D_3)$$

B

(4) 4x2 PRIORITY ENCODER ($I_1 > I_3 > I_0 > I_2$)

I_1	I_3	I_0	I_2	A	B	V
0	0	0	0	X	X	0
1	X	X	X	0	1	1
0	1	X	X	1	1	1
0	0	1	X	0	0	1
0	0	0	1	1	0	1

$I_1 I_3$ \ $I_0 I_2$	00	01	11	10
00	X	1	0	0
01	1	1	1	1
11	0	0	0	0
10	0	0	0	0

For A

~~XXXXXXXXXXXXXXXXXXXX~~

$$A = \bar{I}_1 \bar{I}_0 + \bar{I}_1 I_3$$

$I_1 I_3$ \ $I_0 I_2$	00	01	11	10
00	X	0	0	0
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

For B

$$B = I_3 + I_1$$